A MACHINE-CHECKED PROOF OF SUPERLINEAR CIRCUIT LOWER BOUNDS FOR E^{NP}, ASSUMING THE NEW #SAT ALGORITHMS

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Abstract. By a theorem of Williams, if the Exponential Time Hypothesis fails, then E^{NP} does not have circuits of linear size. In October 2026 OpenAI and Anthropic claimed deterministic subexponential-time algorithms for #k-SAT. OpenAI's revised preprint (25 October 2026) states the quantitative consequence as its Corollary 1.4, proved on paper: some $L \in \mathrm{E}^{\mathrm{NP}}$ has no circuit with at most $c\,n\sqrt{\log n}$ fan-in-two gates for infinitely many n. However, this bound is not machine-checked, and Anthropic's Lean development proves only the linear-size form. We machine-check OpenAI's superlinear circuit lower bound in Lean 4 on top of Anthropic's development, whose formalised engine supplies the rate: `enp_not_size_n_sqrt_log` gives a language $L_0 \in \mathrm{E}^{\mathrm{NP}}$ and $\varepsilon > 0$ (in the proof, $\varepsilon = 1/(4608(qG+1))$ for existential constants q, G) such that for infinitely many n, L_0 has no circuit with at most $\varepsilon\, n\sqrt{\log n}$ gates over the full binary basis, with the axioms `propext`, `Classical.choice` and `Quot.sound` only. On paper, assuming the clause-count bound $2^{C\cdot kM/\sqrt{\log M}}$ of Anthropic's Main Theorem as a Turing machine with engine constant C, we track the constants of Williams' argument: a check circuit of $3s + O(n)$ gates, at most $4S + 1$ clauses after Tseitin's transformation, and any $\varepsilon < 1/(36aC)$, with a the input-length shift. From the weaker bound in OpenAI's Lean development, not the stronger one in its paper, the same argument gives $\varepsilon\, n\sqrt{\log n}/(\log\log n)^2$. We explain why these algorithms give nothing at $n \log n$, nothing for NEXP or NQP, and nothing more from counting for binary gates. The constant ε has no numerical value, the bound holds only infinitely often, and everything is conditional on the claimed refutations, still under review.

1. Introduction

Proving that an explicit function needs large Boolean circuits is one of the central problems of complexity theory, and progress on it is slow. Before 2026 the best lower bound for a language in E^{NP} over the full binary basis was $3.1n - o(n)$, by Li and Yang [10]. Williams' algorithmic method [17] offers another route: a satisfiability algorithm slightly faster than exhaustive search implies a circuit lower bound. In particular, Williams [17, Theorem 6.1] showed that if the Exponential Time Hypothesis (ETH) is false, then E^{NP} does not have circuits of linear size; see also Jahanjou, Miles and Viola [6, Corollary 2(1) with Corollary 3(1′)].

Two manuscripts, by OpenAI [13] and by Anthropic [1] (below [O] and [A]), each claim a deterministic algorithm that counts the satisfying assignments of a k-CNF in subexponential time, and hence a refutation of ETH. Both come with Lean 4 formalisations of their top-level statements. These are *claimed* results, still under review. The Main Theorem of [A] (Theorem 1, made explicit in its Corollary 11) counts the models of a k-CNF with $m \geq 2$ clauses in which $n \geq 4$ variables occur in time

$$\min\left\{2^{C\cdot km/\sqrt{\log m}},\ 2^{C\cdot k^2 n \log\log n/\sqrt{\log n}}\right\}$$

for an absolute constant C. Logarithms are to base 2.

Both manuscripts draw circuit consequences. [O], in its revision of 25 October 2026, states the quantitative one as its Corollary 1.4 [13]: there are $L \in \mathrm{E}^{\mathrm{NP}}$ and $c > 0$ such that L has no circuit with at most $c\,n\sqrt{\log n}$ gates of fan-in at most two, with arbitrary Boolean gates, for infinitely many n. [O] proves it on paper in its Appendix C, by Williams' Theorem 6.1, the local reduction of JMV and Žák's hierarchy theorem, from the constant-density part of its Theorem 1.1. [A] states the linear-size consequence and formalises it in Lean (`ENP.CorCircuits`). Its proof uses only the qualitative statement that, for every $\delta > 0$, 3-SAT is decidable in time $2^{\delta N}$, and it gives "not $O(n)$" with no rate. However, the superlinear bound has no machine-checked proof: [O]'s Corollary 1.4 is proved on paper only, and [A]'s formal corollary does not see the rate.

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1.1. **The question, the naive route and the principle.** We ask for a machine-checked proof of the superlinear bound. At first one might reuse [A]'s formal corollary with a larger size bound. However, its Lean module hard-wires both the size $c_0(m+1)$ and the time $2^{\delta N}$ (Section 4), and the δ-form of the algorithm cannot see a rate at all. Instead, we rely on a simple principle: *in Williams' argument every step except the satisfiability algorithm costs only a linear overhead, so a rate $2^{M/g(M)}$ turns into a size bound $\varepsilon n\, g(n)$.* The chain we track is short. A witness circuit of size $s = \varepsilon\ell\sqrt{\log \ell}$ gives a check circuit of $S \le 3s + O(n)$ gates, Tseitin's transformation gives $M \le 4S + 1$ clauses, and the exponent of the algorithm becomes $36aC\varepsilon n + o(n)$ (Section 3). We now state the model and the two hypotheses, then the results.

1.2. **Model.** A *circuit* is over the full binary basis B_2: each gate computes one of the 16 functions $\{0,1\}^2 \to \{0,1\}$ of two earlier wires. Its *size* is the number of gates; inputs are not counted. Circuits are non-uniform, one per input length. For $s\colon \mathbb{N} \to \mathbb{N}$, a language L *has circuits of size s at length n* if some circuit with at most $s(n)$ gates computes $L \cap \{0,1\}^n$. E^{NP} is the class of languages decided in time $2^{O(n)}$ by a deterministic oracle Turing machine with an oracle for SAT, where queries may have length $2^{O(n)}$.

Hypothesis ($\mathrm{H_A}$)**.** *There are a constant $C \ge 1$ and a deterministic multitape Turing machine that, given a 3-CNF ψ with no repeated clause, with $M \ge 2$ clauses and at least 4 occurring variables, encoded in length L, decides whether ψ is satisfiable within $L^C \cdot 2^{3C\cdot M/\sqrt{\log M}}$ steps.*

This is the m-branch of Corollary 11 of [A] with $k = 3$, stated for Turing machines. We call C here the *engine constant*: the constant of the machine that is actually run. It is not necessarily the C printed in [A]. [A] counts operations in its own model, and turning those into Turing-machine steps changes C by the polynomial overhead of the simulation. In Anthropic's Lean development the rate is available as the machine form `Tied.TopM` (rate $e_M \le G \cdot k \cdot m/\sqrt{\log(m+2)}$, then a polynomial-time machine on a pad of length 2^{e_M}). The engine constant is then determined by G and by the polynomial exponent of that machine.

Hypothesis ($\mathrm{H_O}$)**.** *There are a constant $C_2 \ge 1$ and a deterministic multitape Turing machine that decides the satisfiability of a 3-CNF with $N \ge 16$ variables, encoded in length L, within $L^{C_2} 2^{C_2\rho(N)}$ steps, where $\rho(N) = N(\log\log N)^2/\sqrt{\log N}$.*

This is the form of the theorem `exact_count_and_sat` in OpenAI's Lean development, whose rate function is ρ, up to the base of the logarithms. The printed Theorem 1.1 of [O] is stronger than its Lean form. Its general exponent is $O_k(n\log\log n/\sqrt{\log n})$, with one factor of $\log\log n$, and for $m \le Dn$ it is $O_{k,D}(n/\sqrt{\log n})$. That constant-density bound gives $c\,n\sqrt{\log n}$, and [O] states exactly this as its Corollary 1.4. Corollary 1.2 derives the weaker $(\log\log n)^2$ form from the Lean-formalised statement ($\mathrm{H_O}$) only.

1.3. **What we prove.**

Theorem 1.1 (Main theorem; the statement of [13, Corollary 1.4]; first part machine-checked)**.** *Assume* ($\mathrm{H_A}$)*. There are $\varepsilon > 0$ and $L_0 \in \mathrm{E}^{\mathrm{NP}}$ such that for every N there is an $n \ge N$ for which $L_0 \cap \{0,1\}^n$ is computed by no circuit over B_2 with at most $\varepsilon\, n\sqrt{\log n}$ gates.*

More precisely, there is a constant $a \ge 2$, which depends only on the succinct reduction of Imported Theorem 2.2 applied to one fixed language, such that every ε with $36\,a\,C\,\varepsilon < 1$ works.

Corollary 1.2 (Through OpenAI's Lean-formalised bound)**.** *Assume* ($\mathrm{H_O}$)*. There are $\varepsilon > 0$ and $L_0 \in \mathrm{E}^{\mathrm{NP}}$ such that for infinitely many n, $L_0 \cap \{0,1\}^n$ has no circuit with at most $\varepsilon\, n\sqrt{\log n}/(\log\log n)^2$ gates. Any ε with $3aC_2\varepsilon < 1$ works.*

The n-branch of [A] is a bound in the number of variables, $2^{Ck^2 N\log\log N/\sqrt{\log N}}$. Used in the same way, it gives only $\varepsilon\, n\sqrt{\log n}/\log\log n$. So we use the m-branch.

Remark 1.3 (What the statement does and does not say)*.*

(a) *Infinitely often.* The conclusion is "$\forall N\, \exists n \ge N$". It is not that L_0 fails at a single length. With ε small, $\lfloor \varepsilon n\sqrt{\log n}\rfloor = 0$ for many small n, so a single failing length would be close to vacuous. For a fixed ε, the infinitely-often form cannot be recovered from the single-length form by enlarging the size bound, which is what the linear case does (`ENP.infinitely_often`). We prove it directly by assuming circuits only at lengths $\ge N$. The conclusion does not say that the bound holds for almost every n (Section 6).

(b) *An existential ε.* The constant ε is not explicit, because C is not: [A] and its Lean statement only assert that some C exists. The result therefore does not give $c\,n\sqrt{\log n}$ for every c. That would need a rate $2^{o(M/\sqrt{\log M})}$.

(c) *Strength.* The theorem implies the linear-size statement of [A], and it is strictly more informative: at infinitely many lengths, the lower bound grows faster than every linear function.

1.4. **Contributions.** The conclusion of Theorem 1.1 is the statement of [O]'s Corollary 1.4, proved there on paper from [O]'s own algorithm; we do not claim it. In this paper we machine-check OpenAI's superlinear circuit lower bound. Our contributions are:

- We give a machine-checked proof in Lean of the first part of Theorem 1.1, on top of Anthropic's development, as `enp_not_size_n_sqrt_log`. Its proof uses Anthropic's language `CircuitLowerBound.L0` and the explicit formula $\varepsilon = 1/(4608(qG+1))$ in Anthropic's constants. We quote its statement and list the definitions a reader must check (Section 4).
- We state Turing-machine forms ($\mathrm{H_A}$) and ($\mathrm{H_O}$) of the claimed theorems, and separate the engine constant from the printed one (Section 1).
- We prove Theorem 1.1 on paper from Anthropic's rate ($\mathrm{H_A}$), with the constants tracked: every overhead is linear, and every ε with $36aC\varepsilon < 1$ works (Sections 2 and 3).
- We derive Corollary 1.2, the form with $(\log\log n)^2$, from OpenAI's Lean-formalised bound; OpenAI's paper itself states the stronger $c\,n\sqrt{\log n}$ (Section 3).
- We explain where this route stops: at $n\log n$, below E^{NP}, with counting, almost everywhere, and for restricted classes (Section 6).

What we add to [O]'s Corollary 1.4 is the machine-checked proof, on a different engine and with a different language; the explicit constant bookkeeping on paper; and the discussion of why nothing beyond $n\sqrt{\log n}$ follows. We found our proof without [O]'s corollary: we became aware of it only after the formalisation was complete (4 November 2026), although [O]'s revision is dated 25 October 2026. We claim no independence beyond that. The underlying correspondence is older still: JMV note that their proofs "provide a close correspondence between the running time for satisfiability and the parameters of the circuit class" [6], and Ben-Sasson and Viola's constant-level instance, as quoted there, is: 3-SAT in time c^n with $c < 2^{1/10}$ implies $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(3n)$.

1.5. **Status of each claim.** We separate four kinds of claim. The conclusion of Theorem 1.1 is the statement of [O]'s Corollary 1.4; the labels below concern our proofs. *Machine-checked in Lean*: the first part of Theorem 1.1, as `EthFollowup.Circuits.enp_not_size_n_sqrt_log`, with the variants `enp_not_size_n_sqrt_log_shifted` and `enp_not_hasCircuits_n_sqrt_log`; all with the axioms `propext`, `Classical.choice` and `Quot.sound` only, and a fresh `leanchecker` replay of the dependency closure passed. The Lean theorem has no hypothesis. The role of ($\mathrm{H_A}$) is played by Anthropic's formalised rate engine (the property `Tied.TopM`, which Anthropic proves as `GeneralPieces.thmRunningTime_c_m_tied`), the language is Anthropic's `CircuitLowerBound.L0`, and the proof follows Anthropic's lazy diagonalisation, not the route of Section 3. Its constant is $\varepsilon = 1/(4608(qG+1))$ (Section 4). *Proved on paper only*: the proof of Section 3, including the condition $36aC\varepsilon < 1$ in the second part of Theorem 1.1; Corollary 1.2, which is not claimed as machine-checked; Remarks 1.3 and 3.4; and the discussion of Section 6. *Assumed*: ($\mathrm{H_A}$) and ($\mathrm{H_O}$), and Imported Theorems 2.1 and 2.2, for the proofs on paper. For the Lean theorem, nothing beyond the three axioms; but it stands or falls with Anthropic's development (Section 7). *Checked numerically*: nothing; this note has no computations.

Outline. Section 2 states the imported results, Section 3 gives the proof, Section 4 the Lean development, Section 5 related work, Section 6 what these algorithms cannot give, and Section 7 the limitations.

2. Imported results

In this section we state the two results that the proof on paper takes from the literature, and fix the input-length shift a.

Imported Theorem 2.1 (Nondeterministic time hierarchy, Žák [19])**.** *If t_2 is time-constructible and $t_1(n+1) = o(t_2(n))$, then* $\mathrm{NTIME}[t_1] \subsetneq \mathrm{NTIME}[t_2]$. *In particular, for every $\gamma < 1$ there is a language $L^\star \in \mathrm{NTIME}[2^n] \setminus \mathrm{NTIME}[2^{\gamma n}]$.*

Imported Theorem 2.2 (Local reductions, [6, Theorem 1 and the remark after it])**.** *Let $L \in \mathrm{NTIME}[2^n]$. There are constants d, a_1 and a polynomial-time map sending $x \in \{0,1\}^n$ to a circuit $E_x \colon \{0,1\}^u \to \{0,1\}^{3v+3}$, with $u = n + O(\log n)$ and $v \le a_1 n$, such that the following hold.*

- *$E_x(j)$ lists the three variables (each in $\{0,1\}^v$) and three signs of the j-th clause of a 3-CNF φ_x with 2^u clauses on the variable set $\{0,1\}^v$.*
- *$x \in L$ if and only if φ_x is satisfiable.*
- *Every output bit of E_x is a decision tree of depth $O(1)$ in the bits of j. Hence E_x has at most $d\,n$ gates over B_2.*

JMV state this for algorithms $M(x,y)$ running in time $T \ge 2^{\Omega(n)}$. Their index length is $\ell = \log T + O(\log n)$, and every output bit is a decision tree of depth $O(\log r)$ with $r = n/\log T = O(1)$. Williams' [17, Theorem 6.1] is a *sketch* with the same shape. Its formula comes from a tableau of size $O(2^{3n})$, and its check circuit has "$O(cn)$ wires and $3n + O(1)$ inputs". The explicit count $3s + O(n)$ that we use is the one implicit in JMV's §6 (three copies of the witness circuit and the decision trees) and explicit in Anthropic's Lean module (`StForm`). There the reduction is replaced by a self-contained succinct tableau formula and lazy diagonalisation. We may pad the variable names so that $v = v(n)$ depends only on n and $n \mapsto n + v(n)$ is strictly increasing. Put

$$\ell(n) := n + v(n), \qquad a := \max\{2,\ 1 + a_1\}, \qquad \text{so} \qquad \ell(n) \le a n.$$

3. Proof of Theorem 1.1: every overhead is linear

In this section we define the hard language, give the nondeterministic algorithm, and bound its running time; the last step is where the rate enters. We follow Williams' route [17, Lemma 3.2, Theorem 6.1]. Fix $\varepsilon > 0$ with $36aC\varepsilon < 1$, and fix γ with $36aC\varepsilon < \gamma < 1$. Let $L^\star$ be the language of Imported Theorem 2.1 for this γ, and let $E_x, \varphi_x, u, v, \ell$ be as in Imported Theorem 2.2 for $L = L^\star$. Put

$$s(\ell) := \left\lfloor \varepsilon\, \ell \sqrt{\log \ell} \right\rfloor.$$

3.1. **The language L_0.** For x with φ_x satisfiable, let $w^*(x) \in \{0,1\}^{2^v}$ be its lexicographically first satisfying assignment, indexed by $\{0,1\}^v$. Let L_0 be the set of strings xi with $|x| = n$, $i \in \{0,1\}^{v(n)}$, φ_x satisfiable, and $w^*(x)_i = 1$. Strings whose length is not of the form $\ell(n)$ are not in L_0. Since ℓ is strictly increasing, n and the split xi can be read off from the length.

Lemma 3.1. $L_0 \in \mathrm{E}^{\mathrm{NP}}$.

Proof. This is [17, Lemma 3.2]. On input xi of length $\ell(n)$, write out φ_x by evaluating E_x on all 2^u indices, which takes time $2^{O(n)}$. The language

$$\{\langle \varphi, p \rangle : \varphi \text{ has a satisfying assignment extending the partial assignment } p\}$$

is in NP. Use it to fix the bits of $w^*(x)$ one at a time, in lexicographic order. This takes 2^v queries of length $2^{O(n)}$, after one query that rejects if φ_x is unsatisfiable. Then output bit i. The total time is $2^{O(n)} = 2^{O(\ell)}$. □

3.2. **The algorithm.** Suppose, for a contradiction, that there is an N such that L_0 has circuits of size s at every length $\ell \ge N$. Let n be so large that $\ell(n) \ge N$, and let $x \in \{0,1\}^n$. Consider the following nondeterministic algorithm.

(1) *Guess* a circuit K on $\ell(n)$ inputs with at most $s(\ell(n))$ gates. Hardwire x to get $W(i) := K(xi)$, a circuit on v inputs with at most $s(\ell(n))$ gates (substituting constants does not add gates).

(2) *Build the check circuit* $D \colon \{0,1\}^u \to \{0,1\}$. On input j, compute $E_x(j) = (y_1, \sigma_1, y_2, \sigma_2, y_3, \sigma_3)$. Evaluate the three copies $W(y_1), W(y_2), W(y_3)$. Compute the three literal values $W(y_t) \oplus \neg\sigma_t$ (one gate each), and output 1 if all three are 0 (two gates). So $D(j) = 1$ if and only if clause j of φ_x is violated by the assignment $i \mapsto W(i)$. Its size is

$$S \ \le\ 3\, s(\ell(n)) + d\, n + 5. \tag{3.1}$$

(3) *Tseitin.* Introduce one variable per input and per gate of D. For a gate $g = f(p, q)$, add for each $(\alpha, \beta) \in \{0,1\}^2$ the clause $(p \neq \alpha) \vee (q \neq \beta) \vee (g = f(\alpha, \beta))$, of width at most 3. Add the unit clause on the output gate. Delete tautological and repeated clauses. The result ψ is a 3-CNF with no repeated clause and

$$M \;\le\; 4S + 1 \tag{3.2}$$

clauses. It is satisfiable if and only if $D(j) = 1$ for some j, since the gate variables are forced. The number of variables is $u + S$, so also $M \le 4N_v + 1$, where N_v is the number of variables.

(4) *Decide.* If $M \le 1$ or fewer than 4 variables occur, decide ψ by brute force. Otherwise run the machine of $(\mathrm{H_A})$. *Accept* if and only if ψ is unsatisfiable.

Lemma 3.2 (Correctness)**.** *For $|x| = n$ as above, the algorithm has an accepting branch if and only if $x \in L^\star$.*

Proof. If $x \in L^\star$, then φ_x is satisfiable. Some circuit K of size $\le s(\ell(n))$ computes L_0 at length $\ell(n)$, so on the branch that guesses it, $W(i) = w^*(x)_i$ for every i. That assignment violates no clause, so $D \equiv 0$, ψ is unsatisfiable, and the branch accepts. If $x \notin L^\star$, then every assignment, in particular $i \mapsto W(i)$ for every guessed K, violates some clause j. So $D(j) = 1$, ψ is satisfiable, and every branch rejects. □

3.3. **The arithmetic.**

Lemma 3.3 (Running time)**.** *For all sufficiently large n, every branch halts within $2^{\gamma n}$ steps.*

Proof. Steps (1)–(3) take $\mathrm{poly}(n, s) = 2^{O(\log n)}$ steps, and the encoding length of ψ is $L = \mathrm{poly}(n)$. It remains to bound the exponent $3C\,M/\sqrt{\log M}$ of step (4).

By (3.1), (3.2) and $\ell(n) \le an$,

$$M \;\le\; M^* := 12\,\varepsilon\,a\,n\sqrt{\log(an)} + 4dn + 21.$$

The function $x \mapsto x/\sqrt{\log x}$ is increasing for $x \ge 2$, since its derivative is $(\log x)^{-1/2}\big(1 - \frac{1}{2\ln x}\big) > 0$ there. Also $M^* \ge n$, so $\log M^* \ge \log n$. Hence

$$\frac{3C\,M}{\sqrt{\log M}} \;\le\; \frac{3C\,M^*}{\sqrt{\log n}} \;=\; 36\,a\,C\,\varepsilon\,n\sqrt{\frac{\log(an)}{\log n}} + \frac{3C(4dn+21)}{\sqrt{\log n}} \;=\; 36\,a\,C\,\varepsilon\,n + o(n). \tag{3.3}$$

So the running time is at most $2^{O(\log n)} \cdot L^C \cdot 2^{(36aC\varepsilon + o(1))n}$, which is at most $2^{\gamma n}$ for large n, because $36aC\varepsilon < \gamma$. □

Proof of Theorem 1.1. The algorithm needs to know $s(\ell(n))$, which is computable in time $\mathrm{poly}(n)$. Under the assumption, Lemmas 3.2 and 3.3 decide $L^\star$ correctly within $2^{\gamma n}$ steps on all inputs of all sufficiently large lengths. The finitely many remaining inputs are handled by a table. Hence $L^\star \in \mathrm{NTIME}[2^{\gamma n}]$, which contradicts Imported Theorem 2.1. So for every N there is an $\ell \ge N$, indeed one of the form $\ell(n)$, at which L_0 has no circuit with at most $s(\ell)$ gates. Together with Lemma 3.1, this is the theorem. □

Remark 3.4 (The input-length shift and the constant).

(a) *The shift.* The witness circuit is a circuit for L_0 at length $\ell(n)$, not n. So the check circuit has $3s(\ell(n)) + O(n)$ gates, not "$3s(n) + O(n)$". In JMV's own construction the E^{NP} input is (x, i) of length $2n + O(\log n)$. For $s = \varepsilon n\sqrt{\log n}$ the shift costs only the factor a in the condition on ε.

(b) *Lean normalisation.* Anthropic's module avoids the hierarchy theorem. It takes m to be the input length of its E^{NP} language. The tableau formula there has time 2^{b_t} with $b_t \ge 3m/8$, and the engine is allowed a budget of $2^{m/32}$ inside a total of $2^{m/8}$ (`exists_threshold`). Its check formula has $N_v \le 3|K| + a_0(m+1) + g(e)$ variables and at most $4N_v + 1$ clauses. With $|K| \le \varepsilon(m+1)\sqrt{\log(m+2)}$, the computation (3.3) becomes $36C\varepsilon m(1 + o(1)) + O(m/\sqrt{\log m}) \le m/32 - o(m)$. This holds for $\varepsilon < 1/(1152\,C)$; for instance $\varepsilon = 1/(1200C)$.

(c) *Which C.* In either normalisation, C is the effective engine constant of $(\mathrm{H_A})$. In Lean this is G together with the polynomial exponent of the machine on the pad, and possibly the exponent of the step bound in `PaddedDescent.exists_tm'`. It is fixed before the formula table is chosen, so ε is still an absolute constant.

Proof of Corollary 1.2. Take $s(\ell) = \lfloor \varepsilon \ell \sqrt{\log \ell}/(\log\log \ell)^2 \rfloor$ and $3aC_2\varepsilon < \gamma < 1$. Run the same algorithm with the machine of ($\mathrm{H_O}$) in step (4). Now $N_v \le u + S \le N^* := 3\varepsilon a n\sqrt{\log(an)}/(\log\log n)^2 + (d+2)n$ for large n. The function ρ is increasing for large arguments. We have $\log N^* = \log n + O(\log\log n)$, and so $\log\log N^* = \log\log n + o(1)$. Therefore

$$C_2\,\rho(N_v) \le C_2\,\rho(N^*) = 3aC_2\varepsilon\, n(1+o(1)) + O\!\left(\frac{n(\log\log n)^2}{\sqrt{\log n}}\right) \le (\gamma - o(1))\, n,$$

and the rest is unchanged. □

4. The Lean development

In this section we say where the formal proof is, quote its statement exactly, list the definitions a reader must check, and state the trusted base.

Anthropic's linear form. Anthropic's Lean development [1] proves the linear-size form,

$$\texttt{ENP.CorCircuits}: \ \exists L,\ \texttt{InENP}\ L \wedge \neg\, \texttt{HasLinearCircuits}\ L,$$

together with the infinitely-often variant `CorCircuitsInf`. Its circuits are over B_2, and size is the number of gates. E^{NP} is formalised with Mathlib `FinTM2` oracle machines. The proof follows the route above, using the δ-form of the engine (time $c \cdot 2^{\delta N}$ for every $\delta > 0$) and lazy diagonalisation in place of Imported Theorem 2.1. The statement `HasCircuits` $L\ s$ is already parametric in s, but the module hard-wires both the size $c_0(m+1)$ and the time $2^{\delta N}$.

Where the proof is. Theorem 1.1 (first part) is machine-checked in Lean 4 on top of that module. The proof is in the public repository `https://github.com/xennygrimmato/eth-hamiltonicity` [15], at commit `9e5dfc8`, in the directory `lean-bridge-anthropic/bridge/EthFollowup/Circuits/`: five files, 707 lines (Table 1). It builds on Anthropic's development at the commit pinned in the repository (`anthropics/formal-math`, folder `eth`, commit `16a5ce4`, Lean v4.33.1).

file	content	lines
`Main.lean`	`enp_not_size_n_sqrt_log` and its two variants; the size function and the interface statement of the algorithm	128
`LowerBound.lean`	the lower-bound argument for a fixed size function, infinitely often (`no_circuits_io`)	93
`Engine.lean`	the rate engine `Tied.TopM` packaged as one padded polynomial-time function	202
`FormCount.lean`	the clause bound $M \le 4N_v + 1$ for the Tseitin formula	50
`Arith.lean`	the threshold arithmetic, with $\varepsilon = 1/(4608(qG+1))$	234
total		707

Table 1. The five files of the formal proof, with line counts (`wc -l`).

The statement. The theorem is in `Main.lean`, namespace `EthFollowup.Circuits`, with `open ENP`. We copy it, rendering Unicode symbols as mathematics:

```
theorem enp_not_size_n_sqrt_log : ∃ (L : Lang) (ε : ℝ), 0 < ε ∧ InENP L ∧
    ∀ N : ℕ, ∃ n : ℕ, N ≤ n ∧ ¬DecidedAt L ⌊ε * n * Real.sqrt (Real.logb 2 n)⌋₊ n
```

Here $\lfloor\cdot\rfloor_+$ is `Nat.floor`, `Real.logb 2` is $\log_2$, and the natural number n is cast to $\mathbb{R}$ inside the floor. The variant `enp_not_size_n_sqrt_log_shifted` has the size $\lfloor \varepsilon(n+1)\sqrt{\log_2(n+2)}\rfloor$ in place of $\lfloor \varepsilon n\sqrt{\log_2 n}\rfloor$, and `enp_not_hasCircuits_n_sqrt_log` concludes $\neg$ `HasCircuits` L $(n \mapsto \lfloor \varepsilon n\sqrt{\log_2 n}\rfloor)$, the negation of the module's `HasCircuits`.

The definitions a reader must check. `Lang`, `InENP` and `DecidedAt` are Anthropic's own definitions, in namespace `ENP`, from the statement files `eth/proof/statements/CircuitLowerBoundStatement.lean` and `eth/proof/statements/CircuitLowerBoundInfinitelyOftenStatement.lean` of [1]. The first imports Mathlib only, the second imports the first, and both contain definitions only. We copy what the statement unfolds to, omitting docstrings and rendering Unicode as mathematics:

```
abbrev Lang : Type := List Bool → Prop
structure Gate where
  op : Bool → Bool → Bool
```

```
  in₁ : ℕ
  in₂ : ℕ
abbrev Circuit : Type := List Gate
def wires (x : List Bool) (C : Circuit) : List Bool :=
  C.foldl (fun w G => w ++ [G.op (w.getD G.in₁ false) (w.getD G.in₂ false)]) x
def eval (C : Circuit) (x : List Bool) : Bool := (wires x C).getLastD false
def DecidedAt (L : Lang) (s n : ℕ) : Prop :=
  ∃ C : Circuit, C.length ≤ s ∧ ∀ x : List Bool, x.length = n → (eval C x = true ↔ L x)
def HasCircuits (L : Lang) (s : ℕ → ℕ) : Prop :=
  ∀ n : ℕ, ∃ C : Circuit, C.length ≤ s n ∧ ∀ x : List Bool, x.length = n → (eval C x = true ↔ L x)
abbrev Cnf : Type := List (List (ℕ × Bool))
def Sat (φ : Cnf) : Prop := ∃ a : ℕ → Bool, ∀ C ∈ φ, ∃ l ∈ C, a l.1 = l.2
def encLit (l : ℕ × Bool) : List Bool := true :: (List.replicate l.1 true ++ [false, l.2])
def encCnf (φ : Cnf) : List Bool :=
  φ.flatMap (fun C => C.flatMap encLit ++ [false, true]) ++ [false, false]
def SAT : Lang := fun w => ∃ φ : Cnf, w = encCnf φ ∧ Sat φ
structure OTM where
  tm : FinTM2
  inp : tm.Γ tm.k₀ ≃ Bool
  out : tm.Γ tm.k₁ ≃ Bool
  q : tm.K
  qry : tm.Γ q ≃ Bool
  ask : tm.Λ
  yes : tm.Λ
  no : tm.Λ
open Classical in
noncomputable def OTM.step (M : OTM) (O : Lang) (c : M.tm.Cfg) : Option M.tm.Cfg :=
  if c.l = some M.ask then
    some ⟨some (if O ((c.stk M.q).map M.qry) then M.yes else M.no), c.var,
      Function.update c.stk M.q []⟩
  else M.tm.step c
def OTM.Outputs (M : OTM) (O : Lang) (x : List Bool) (b : Bool) (T : ℕ) : Prop :=
  Nonempty (StateTransition.EvalsToInTime (M.step O) (initList M.tm (x.map M.inp.symm))
    (some (haltList M.tm [M.out.symm b])) T)
def InENP (L : Lang) : Prop :=
  ∃ (M : OTM) (c : ℕ), (∀ k, Finite (M.tm.Γ k)) ∧
    ∀ x : List Bool, ∃ b : Bool, (b = true ↔ L x) ∧ M.Outputs SAT x b (2 ^ (c * (x.length + 1)))
```

In words: `DecidedAt L s n` says that some circuit with at most s gates agrees with L on all words of length n. A gate is any of the 16 functions of two wires; the wires are the input bits followed by the gate values; the output is the last wire; a gate that names a wire that does not exist yet reads 0 there. `InENP L` says that one oracle machine with finite stack alphabets and one constant c output, with the oracle `SAT` for satisfiability of CNFs, the bit "$x \in L$" on every input x within $2^{c(|x|+1)}$ steps. A query costs one step, and the machine pushes it bit by bit, so queries have length at most $2^{c(|x|+1)}$. The remaining names are Mathlib's: `Turing.FinTM2` (with its step function `Turing.TM2.step`), `Turing.initList`, `Turing.haltList`, `StateTransition.EvalsToInTime`, `Nat.floor`, `Real.logb` and `Real.sqrt`. The language L of the proof is Anthropic's `CircuitLowerBound.L0`, with membership `CircuitLowerBound.stENP`; a reader need not check its definition, because the statement quantifies over L.

How the proof goes. The proof threads the size function through Anthropic's module. Three points needed more than arithmetic. First, the rate solver is packaged as one padded polynomial-time function and turned into a Turing machine by Anthropic's own padded-host lemma (`PaddedDescent.exists_tm_padded`), so no polynomial bound on the machine's states is needed. Second, the solver counts over the occurring variables only; the proof compares that count with zero directly. Third, the infinitely-often form is obtained by threading the threshold N through the module's diagonalisation (`no_circuits_io`). The constant is $\varepsilon = 1/(4608(qG+1))$, where q is the host's polynomial degree and G the constant of the rate solver; it is existential because q and G are.

Trusted base and axioms. `#print axioms` reports `propext`, `Classical.choice` and `Quot.sound` for `enp_not_size_n_sqrt_log` and `enp_not_hasCircuits_n_sqrt_log`, and a fresh `leanchecker` replay of the full dependency closure passed. What must be trusted is the Lean kernel, the Mathlib definitions named above, and Anthropic's definitions quoted above. Anthropic's development and Mathlib's proofs are checked by

the kernel with ours; they are part of the checked closure, not of the trusted base of definitions. But the soundness of the whole rests on them building as a sound development (Section 7).
What is not formalised. Corollary 1.2 (the OpenAI form) is proved on paper only and is not claimed as machine-checked. The route of Section 3 through Imported Theorems 2.1 and 2.2, the condition $36aC\varepsilon < 1$, and Remark 3.4 are not formalised; Lean proves the first part of Theorem 1.1 with its own constant.

5. Related work

We organise related work by idea: explicit lower bounds for general circuits, near-maximum bounds above E^{NP}, the algorithmic method, the claimed refutations with their formal proofs, and the extension to larger gate bases. The statements below were checked against the primary sources.
Explicit lower bounds for E^{NP} and P over B_2. Before 2026 the best lower bound for a language in E^{NP} was $3.1n - o(n)$, by Li and Yang [10]. They prove it for affine dispersers of sublinear dimension, which are constructible in polynomial time, so the bound holds already in P. It improves the $(3 + \frac{1}{86})n - o(n)$ of Find, Golovnev, Hirsch and Kulikov [4], which in turn improves the $3n - o(n)$ of Blum. Williams wrote in 2010 that "linear size circuit lower bounds are not known for this class" [17], that is, no proof that linear size does not suffice. In contrast, Theorem 1.1 is superlinear, but conditional, and it holds only infinitely often.
Near-maximum and fixed-polynomial bounds above E^{NP}. Chen, Hirahara and Ren [2] prove that $\mathrm{S_2E}/_1 \not\subseteq \mathrm{SIZE}[2^n/n]$. This gives the same near-maximum bounds for $\Sigma_2\mathrm{E}$, $(\Sigma_2\mathrm{E} \cap \Pi_2\mathrm{E})/_1$ and $\mathrm{ZPE}^{\mathrm{NP}}/_1$. The almost-everywhere, advice-free bound $\mathrm{S_2E} \not\subseteq$ i.o.-$\mathrm{SIZE}[2^n/n]$ is due to Li [9]. $\mathrm{E}^{\mathrm{NP}} \subseteq \mathrm{S_2E}$, so these classes lie above the one treated here. Kannan [7] showed that for each k, $\Sigma_2\mathrm{P} \cap \Pi_2\mathrm{P} \not\subseteq \mathrm{SIZE}(n^k)$. Whether $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(n^k)$ for some fixed $k > 1$ is open. Before the refutations, so was $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$. In contrast, our class is E^{NP} itself, and our bound is far below these sizes.
The algorithmic method. Williams [17, Lemma 3.2, Theorem 6.1] introduced the route we follow, and his ACC lower bounds [18] and those of Murray and Williams [12] exploit counting or fast evaluation, for polynomial-size circuits. Jahanjou, Miles and Viola [6] made the reduction local, which is what keeps the check circuit linear, and remark on the close correspondence between running times and circuit parameters; Ben-Sasson and Viola's instance $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(3n)$ is quoted there. Below E^{NP} one needs an easy-witness lemma [5, 12], and JMV raise the strengthening of [5] to linear-size circuits as an open problem. Chen, Lyu and Williams [3] obtain almost-everywhere lower bounds for E^{NP} against ACC^0. No superlinear lower bound for NEXP follows from our argument. In contrast to these works, we substitute a subexponential rate and keep every overhead linear, which gives a superlinear bound.
The claimed refutations and their circuit corollaries. OpenAI [13] and [A] [1] claim deterministic subexponential #SAT algorithms, with Lean developments, under review. Closest to our work is [O]'s Corollary 1.4 [13, Corollary 1.4], which states the bound of Theorem 1.1, $c\,n\sqrt{\log n}$ for infinitely many n with arbitrary fan-in-two gates, and proves it on paper by the same route: Williams' witness circuits, the local reduction of JMV, Žák's hierarchy theorem, and the constant-density part of [O]'s Theorem 1.1. [O]'s Lean development formalises the counting theorem with the weaker exponent ρ, not the corollary. [A] formalises the linear-size consequence (Section 4). Our earlier work [14] counts Hamiltonian cycles on top of [A]'s development. We use Lean 4 [8] and Mathlib [11]. In contrast to [O]'s corollary, our proof of the bound is machine-checked, from Anthropic's engine rather than OpenAI's; in contrast to [A]'s formal corollary, the Lean theorem here depends on the rate of the engine, not only on its δ-form.
Larger gate bases. Wu and Wang [16] extend [O]'s binary-gate bound to richer bases. From the variable branch of [A]'s Theorem 1 they obtain a language in E^{NP} that, at infinitely many input lengths m, needs more than $c\,m\sqrt{\log m}/\log\log m$ gates over arbitrary-fan-in AND and OR with binary XOR [16, Theorem 1]. From an extension-ring form of Theorem D.1 of [O] they obtain $c\,m\sqrt{\log m}$ gates infinitely often for fan-in-two circuits that may also use arbitrary-fan-in XOR and symmetric gates within a budget of δm, where an XOR gate costs one and a symmetric gate of arity d costs $\log_2(d+1)$ [16, Theorem 2]. Both are proved by a quantitative form of the same local-reduction transfer. Their first bound agrees with our observation after Corollary 1.2: they need the variable branch of [A], which loses a factor $\log\log m$, because long AND and OR gates make the clause count polynomial. Nothing in [16] contradicts a statement here. In contrast to their work, we stay with fan-in-two gates over B_2, where the clause branch applies, and what we add is a machine-checked proof; their bases are larger, and they report no formalisation of their proofs.

6. What these algorithms cannot give

In this section we explain where the route of Section 3 stops with the claimed rates.

- *Beyond $n\sqrt{\log n}$.* The check circuit and Tseitin's transformation are already linear. So the only lever is the rate: an algorithm in time $2^{M/g(M)}$ converts into $s = \varepsilon n\, g(n)$. Size $n \log n$ would need 3-SAT in $2^{O(M/\log M)}$, and size $n^{1+\delta}$ would need $2^{M^{1-\delta'}}$. Neither manuscript claims such a rate.
- NEXP *and* NQP. Moving below E^{NP} needs an easy-witness lemma. Those of Impagliazzo, Kabanets and Wigderson [5] and of Murray and Williams [12] work for polynomial size. The second loses a polynomial in the size. After the loss, one needs satisfiability of $\mathrm{poly}(n)$-size circuits in time $2^n/n^{\omega(1)}$, and $2^{O(s/\sqrt{\log s})}$ with $s = n^k$ gives nothing. Linear-size easy witnesses are JMV's open problem.
- *Counting adds nothing over B_2.* The NP oracle of E^{NP} is used only to fix the lexicographically first witness, and the final check is a decision question. The counting form of the algorithms is not used. The lower bounds of the algorithmic method that do exploit counting or fast evaluation, such as those against ACC of Williams [18] and Murray–Williams [12], concern polynomial-size circuits and run into the previous item. Over larger bases counting does help at near-linear size: Wu and Wang [16] write the check for arbitrary-fan-in AND and OR gates as a difference of two model counts, so that a long gate costs one extra variable rather than one per input (Section 5).
- *Almost everywhere.* Chen, Lyu and Williams [3] obtain almost-everywhere lower bounds for E^{NP} (against ACC^0) from non-trivial algorithms. We have not checked whether their machinery keeps the check circuit at $O(s)$ gates, so that it would give an almost-everywhere version of Theorem 1.1. Their reductions involve PCPs, whose polylogarithmic overhead would absorb the $\sqrt{\log n}$.
- *Restricted classes.* The algorithms are CNF algorithms, and the reduction goes through general circuits. The implied bounds for formulas or for log-depth circuits are no better than what Theorem 1.1 gives, and for formulas they are weaker than known bounds.

7. Limitations

- *The result stands or falls with the claimed refutations.* The proofs on paper are conditional on ($\mathrm{H_A}$), or on ($\mathrm{H_O}$) for Corollary 1.2. These are Turing-machine forms of the claimed theorems of [A] and [O]. Expert review of the manuscripts is pending, and if the hypotheses fail, the results lose their proofs. The Lean theorem rests on Anthropic's development, built from source at the pinned commit, being sound.
- *No human has checked the work.* Our proofs on paper and the Lean code were written with AI assistance and have not yet been checked by a human.
- *The statement is not ours.* The conclusion of Theorem 1.1 is [O]'s Corollary 1.4, proved there on paper. Our contribution is a machine-checked proof on Anthropic's development and the bookkeeping on paper.
- *Only the first part of Theorem 1.1 is machine-checked.* The condition $36aC\varepsilon < 1$, the route through Imported Theorems 2.1 and 2.2, and Corollary 1.2 are proved on paper only.
- *ε is not explicit.* It depends on the effective engine constant C, which is existential in [A] and in its Lean statement, and on the constant a of the reduction. No numerical value of ε is claimed. The bound gives no meaningful statement at any concrete input length.
- *Infinitely often only.* The lower bound holds at infinitely many lengths, not at almost every length (Section 6).
- *Imported reduction.* Imported Theorem 2.2 is used as stated by JMV, with the linear gate count read off from depth-$O(1)$ decision trees. Williams' Theorem 6.1, which we follow, is itself a sketch. The self-contained version is Anthropic's Lean module, for the linear case.
- *Literature.* The statements in Section 5 and all citations were checked against the primary sources (publisher records, arXiv, ECCC, the author's version of [17]) on 4 November 2026. We missed [O]'s Corollary 1.4 until after the formalisation was complete. A novelty search of arXiv and Hexagon on 5 November 2026 found the parallel work of Wu and Wang [16] on larger gate bases (Section 5), and no other statement of Theorem 1.1.

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